ZZ-T-831D
July 20, 1978
SUPERSEDING
Fed. Spec. ZZ-T-831C
September 19, 1967

FEDERAL SPECIFICATION

TUBING; RUBBER AND PLASTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rubber and plastic tubing for laboratory and special use.

1.2 Classification. The tubing shall be of the following types and classes, as specified (see 6.2 and 6.3):

Type I - Commercial rubber tubing.
  Class 1 - Light wall.
  Class 2 - Heavy wall.
  Class 3 - Pressure.
  Class 4 - Gooch.
Type II - Synthetic rubber tubing.
  Class 1 - Light wall.
  Class 2 - Heavy wall.
  Class 5 - Vacuum connections.
Type III - Natural rubber compounded tubing.
Type IV - Latex tubing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-2[illegible]
Check the source to verify that this is the current version before use.

Type V - Laboratory pure gum, natural rubber, or the synthetic equivalent (CIS 1-4 polyisoprene) tubing.
Class 1 - Light wall.
Class 2 - Heavy wall.
Class 3 - Pressure.
Class 4 - Gooch.
Type VI - Plastic tubing.
Class 6 - General use.
Class 7 - Pressure and vacuum.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Specification:

MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (including Tubing); and Fittings, Nozzles, and Strainers.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging for Supplies and Equipment.

(Copies for Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (ASTM) Standards:

D 412 - Tension Testing of Vulcanized Rubber.
D 471 - Change in Properties of Elastometric Vulcanizates Resulting from Immersion in Liquids.
D 543 - Standard Method of Test for Resistance of Plastics to Chemical Reagents.
D 573 - Accelerated Aging of Vulcanized Rubber by the Oven Method.
D 792 - Specific Gravity and Density of Plastics by Displacement.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association (NFPA):

No. 56A - Inhalation Anesthetics (Flammable and Nonflammable).

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

3. REQUIREMENTS

3.1 Description. The tubing shall be of uniform cross section and the surface shall be smooth.

3.2 Material. Material shall be as specified herein (see 6.3).

3.3 Physical properties. The physical properties of the tubing shall be as shown in table I:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

TABLE I. Physical properties.

Property	Type I	Type II	Type III	Type IV	Type V	Type VI
Specific gravity (max.)	1.40	- - -	- - -	- - -	0.97	- - -
Tensile strength: Initial (psi) (min.)	2400	1100	1200	3500	3000	2000
After air heat aging (7 days) (percent of initial) (min.)	75	75	75	75 1/	75	- - -
After 14 days immersion in a 3 percent salt solution at 82° F (28° C) (psi) (min.)	- - -	- - -	- - -	- - -	- - -	2000
Ultimate elongation: Initial (percent) (min.)	650	450	400	750	700	300
After 14 days immersion in a 3 percent salt solution at 82° F (28° C) (percent) (min.)	- - -	- - -	- - -	- - -	- - -	300
Tensile stress at 100 percent elongation (psi) (max.)	150	200	300	- - -	125	800

1/ Except that type IV tubing, .125-inch inside diameter shall retain a minimum of 65 percent of its original strength.

3.4 Electrical conductivity. When specified (see 6.2), rubber tubing shall be electrically conductive. The maximum resistance value over a 3-foot section of electrically conductive tubing shall be 1.0 megohm before and after exposure to oxygen aging, liquid ethyl ether, oil immersion, and flexing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

Color. The color of types I, II, III, and VI tubing shall be amber, black, red, white, or clear; and type V tubing shall be translucent or black, as specified (see 6.2). Unless otherwise specified (see 6.2), the color of the type IV tubing shall be translucent amber.

3.6 Identification marking. Tubing shall be marked or labeled with nomenclature, type, class, dimensions, whether electrically conductive (when applicable), and date of manufacture.

3.7 Type I, commercial rubber tubing. Type I tubing shall be natural rubber, synthetic rubber, or a mixture of the two.

3.7.1 Dimensions. The nominal size of the tubing shall be as specified (see 6.2) and shall be in accordance with table II. Allowable tolerance shall be plus or minus .016 inch for inside diameter and wall thickness. Length of the tubing shall be as specified (see 6.2).

.8 Type II, synthetic rubber tubing. Type II tubing shall be compounded from synthetic rubber.

3.8.1 Dimensions. The nominal size of the tubing and vacuum connections shall be as specified (see 6.2) and shall be in accordance with table III. Allowable tolerance shall be plus or minus .016 inch for inside diameter and wall thickness. Length shall be as specified (see 6.2).

Type III, natural rubber compounded tubing. Type III shall be compounded from natural rubber.

3.9.1 Dimensions. The inside diameter and wall thickness shall be .062 inch. Allowable tolerance shall be .016 inch for inside diameter and wall thickness. Length of the tubing shall be 3 feet plus 1 inch minus 0.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

TABLE II. Dimensions, types I and V tubing.

Nominal size	Class 1, light wall		Class 2, heavy wall		Class 3, pressure		Class 4, Gooch
	Inside diameter	Wall thickness	Inside diameter	Wall thickness	Inside diameter	Wall thickness	Inside width measured flat
(inch)	(inch)	(inch)	(inch)	(inch)	(inch)	(inch)	(inches)
1/8	.125	.031	.125	.062	.125	.188	1.250
3/16	.188	.047	.188	.094	.188	.188	1.500
1/4	.250	.062	.250	.094	.250	.188	1.750
5/16	.312	.062	.312	.094	.312	.188	
3/8	.375	.062	.375	.125	.375	.250	
1/2	.500	.094	.500	.125	.500	.250	
5/8	-	-	.625	.125	-	-	
3/4	-	-	.750	.125	-	-	
1	-	-	1.000	.125	-	-	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:2
Check the source to verify that this is the current version before

TABLE III. Dimensions, type II tubing.

Nominal size	Class 1, light wall		Class 2, heavy wall		Class 5, vacuum connections	
	Inside diameter	Wall thickness	Inside diameter	Wall thickness	Inside diameter	Wall thickness
(inch)	(inch)	(inch)	(inch)	(inch)	(inch)	(inch)
1/8	.125	.047	.125	.094	.188	.188
3/16	.188	.047	.188	.094	.250	.188
1/4	.250	.062	.250	.125	-	-
5/16	.312	.062	.312	.125	-	-
3/8	.375	.062	.375	.125	-	-
1/2	.500	.062	.500	.125	-	-
5/8	.625	.062	.625	.125	-	-
3/4	-	-	.750	.125	-	-
1	-	-	1.000	.125	-	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

3.10 Type IV latex rubber tubing. Type IV tubing shall be liquid natural rubber latex. The nominal size of the tubing shall be as specified (see 6.2) and shall be in accordance with table IV. The length of the tubing shall be as specified (see 6.2).

TABLE IV. Dimensions and tolerance, type IV tubing, inches.

Nominal size	Inside diameter	Tolerance plus or minus	Wall thickness	Tolerance plus or minus
1/8	.125	.016	.031	.008
3/16	.188	.016	.094	.016
1/4	.250	.016	.062	.016
1/4	.250	.016	.094	.016
5/16	.312	.016	.062	.016

3.11 Type V, laboratory pure gum, natural rubber, or synthetic (CIS 1-4 polyisoprene) tubing. Type V tubing shall be natural rubber or synthetic equivalent (CIS 1-4 polyisoprene).

3.11.1 Dimensions. The nominal size of the tubing shall be as specified (see 6.2) and shall be in accordance with table II. The length of the tubing shall be as specified (see 6.2). Allowable tolerance of tubing up to 12 feet in length shall be plus or minus .016 inch inside diameter and wall thickness. When lengths greater than 12 feet are specified, the allowable tolerance of the inside diameter and wall thickness shall be plus or minus .016 inch or 10 percent, whichever is greater.

3.12 Type VI, plastic tubing. Type VI plastic tubing shall contain polyvinyl chloride as the base material.

3.12.1 Physical properties. In addition to the physical properties specified in table I, type VI tubing shall meet the following requirements.

3.12.1.1 Swelling. There shall be not more than 2 percent swelling of the tubing when immersed in a salt solution for 14 days.

3.12.1.2 Resistance to cold. There shall be no signs of cracking when the tubing is bent back upon itself at -1° ± 2° F (-18° ± 1° C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

12.1.3 Flammability. Type VI tubing shall be self extinguishing within 5 seconds.

3.12.2 Dimensions. The nominal size, general use, and pressure and vacuum connection tubing shall be as specified (see 6.2) in accordance with table V. Allowable tolerance shall be plus or minus .016 inch for inside diameter and wall thickness. The length of the tubing shall be as specified (see 6.2).

TABLE V. Dimensions, type VI tubing.

Nominal size	Class 6, general use		Class 7, pressure and vacuum connections	
	Inside diameter	Wall thickness	Inside diameter	Wall thickness
(inch)	(inch)	(inch)	(inch)	(inch)
1/8	.125	.094	.125	.250
3/16	.188	.094	.188	.250
1/4	.250	.125	.250	.250
5/16	.312	.125	-	-
3/8	.375	.125	-	-
1/2	.500	.125	-	-
5/8	.625	.125	-	-
3/4	.750	.125	-	-
1	1.000	.125	-	-

3.13 Workmanship. The tubing shall be essentially free from oil, grease, dirt, sand, fins, pits, porosity, blowholes, tack, bleeding, blooming, and foreign matter. The tubing shall be clean and smooth, except that impressions left by a fine fabric used for wrapping the tube before vulcanization or other slight extrusion marks shall not be cause for rejection. Dusting agents (metallic silicates or stearates) on the surface of latex tubing shall not be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of 5,000 feet, or fraction thereof, of one type, one class, one color, and one size of tubing produced within a 30-day period.

4.3.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. The unit or product for visual examination shall be 3 linear feet. The samples for tests shall consist of sufficient quantity of tubing to perform all tests specified herein. When the size of tubing is impractical for tests, separate specimens shall be prepared of the same material, and of suitable size and shape.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.4 Tests. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.4.2.1 through 4.4.2.3.3, as applicable. Failure of any test shall be cause for rejection of the lot.

4.4 Inspection procedure.

4.4.1 Examination. The tubing shall be examined as specified herein for the following major defects:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

101. Materials not as specified.
102. Color not as specified.
103. Marking not as specified.
104. Dimensions not as specified.
105. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Physical properties. The tubing shall be tested as specified in table VI. Nonconformance to any requirement specified in table I shall constitute failure of the test.

TABLE VI. Tubing test methods.

Property	ASTM
Specific gravity	D 792
Tensile strength:	
Initial	D 412
Airheat test	D 573
Immersion (3 percent salt solution)	D 543
Ultimate elongation:	
Initial	D 412
Immersion (3 percent salt solution)	D 543
Tensile stress at 100 percent elongation	D 412

4.4.2.2 Electrical conductivity. Two 3-foot lengths of the electrical conductive rubber tubing shall be tested in accordance with the applicable procedures specified in NFPA No. 56A except that the electrodes shall be attached at or near both ends of the tubing. Nonconformance to 3.4 shall constitute failure of the test. Proof of acceptability of tubing shall consist of either a listing by Underwriters Laboratories, Inc., or a laboratory report.

4.4.2.2.1 Oxygen aging. The two 3-foot lengths of electrically conductive rubber tubing shall be aged as specified in ASTM D 572 for a period of 94 $\pm$ 1/2 hours.

4.4.2.2.2 Liquid ethyl ether. The two 3-foot lengths of electrically conductive rubber tubing shall be maintained for 5 hours full of ethyl ether in liquid phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

4.2.2.3 Oil immersion. The two 3-foot lengths of electrically conductive rubber tubing shall be immersed in oil conforming to ASTM D 471, No. 2, for 120 hours at a temperature of 73.4° ± 3.6° F (23° ± 2° C).

4.4.2.2.4 Flexing. The two 3-foot lengths of electrically conductive rubber tubing shall be passed back and forth over a pulley 1.750 inches in diameter. One end of the tubing shall be fixed vertically with the end facing down. The tubing shall pass over the pulley so that the movable end will hang vertically downward. A 5-pound weight shall be suspended from the second end. The pulley shall be caused to raise and lower vertically for a distance of 2 feet at a rate of 50 times (25 cycles) per minute for 2,500 cycles.

4.4.2.2.5 Rejection criteria. After exposure and flexing as specified in 4.4.2.2.1 through 4.4.2.2.4, the tubing shall again be tested as specified in 4.4.2.2. Nonconformance to 3.4 shall constitute failure of the test.

4.4.2.3 Type VI, plastic tubing. In addition to the applicable tests specified in 4.4.2.1, type VI tubing shall be subjected to the following tests.

4.4.2.3.1 Swelling. The tubing shall be immersed in a 3 percent sodium chloride solution at a temperature of 82° ± 2° F (28° ± 1° C) for a period of not less than 14 days. The volume of the tubing before and after immersion shall be determined. Evidence of swelling in excess of 2 percent shall constitute failure of this test.

4.4.2.3.2 Resistance to cold. The tubing shall be exposed to an ice salt mixture at a temperature of minus 1° ± 2° F (-18° ± 1° C) for a period of 4 hours. At the end of the exposure period, the tubing shall be removed from the low temperature and quickly bent back on itself. Evidence of cracking shall constitute failure of the test.

4.4.2.3.3 Flammability. The tubing shall be exposed to an open flame for 15 seconds and then removed from the flame. Continued flaming 5 seconds after removal from the open flame shall constitute failure of the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

Inspection of packaging. The preservation, packing, and marking for level A or level B shall be examined to determine compliance with the Quality Assurance Provisions of MIL-P-775. The commercial preservation, packing, and marking shall be examined to determine compliance with MIL-STD-1188.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Tubing shall be preserved in accordance with level A preservation and packaging requirements of MIL-P-775.

5.1.2 Commercial. Tubing shall be preserved in accordance with MIL-STD-1188.

Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Tubing shall be packed in accordance with the level A packing requirements of MIL-P-775.

5.2.2 Level B. Tubing shall be packed in accordance with the level B packing requirements of MIL-P-775.

Commercial. Tubing shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial packaging shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tubing is intended for use in medical and laboratory procedures and special use where softness and high purity are required and when electrical conductivity may be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

2 Ordering data. Purchasers should select the preferred options itted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class tubing required (see 1.2).
(c) When electrical conductive tubing is required (see 3.4).
(d) Color of tubing required (see 3.5).
(e) Nominal size and length of tubing required (see 3.7.1, 3.8.1, 3.10, 3.11.1, and 3.12.2).
(f) Degree of preservation and packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged than recycled material be used, when practical, as long as it meets the requirements of this specification.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Army - MI, EA, AV, AR, MD
Navy - OS
- CS

User activities:
Army - AT
Navy - MS, AS

Preparing activity:
Army - ME

Civil Agency Coordinating Activities:
GSA - FSS
VA - DMS
HEW - FED, NIH

Project 4720-0494

Orders for this publication are to be placed with General Services dministration, acting as an agent for the Superintendent of Documents. ee Section 2 of this specification to obtain extra copies and other ocuments referenced herein. Price 90 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. Contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

ZZ-T-381D, Tubing; Rubber and Plastic

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:28Z
Check the source to verify that this is the current version before use.